Wei-Jin Huang

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Research Summary

I am a Ph.D. student at Sun Yat-sen University, advised by Prof. Wei-Shi Zheng, working on humanoid learning and video action understanding. My research builds physically grounded systems that learn whole-body human–humanoid interaction from human demonstrations and that understand fine-grained human actions in video. I have published as (co-)first author at CVPR, ICCV, ECCV, and ACM MM, and I aim to advance embodied intelligence through rigorous, impactful research.

Internship / Collaboration Interests

Embodied interaction and continual learning are key challenges for general intelligence. I study cross-task robot policy generalization: understanding novel goals and composing learned skills for unseen tasks. I aim to combine Vision-Language-Action (VLA) models, World Action Models (WAMs), agentic planning, continual learning, and egocentric human data to improve compositional generalization, rapid adaptation, and failure recovery.

Education

Ph.D., Computer Science and Technology – Sun Yat-sen University (SYSU), Guangzhou, China — 2024.09 – Present
Specialization: Humanoid Learning, Video Action Understanding

B.E., Network Engineering – South China University of Technology (SCUT), Guangzhou, China — 2020.09 – 2024.07
Average grade ranking 3/66

Selected Publications

1. **ChoreoPlan: Hybrid Phrase Planning and Execution-Grounded Selection for Music-to-Humanoid Dance**
   Wei-Jin Huang, Jianhong Fan, Hao Huang, Zhi-Wei Xia, Jun-Yi Deng, Yuan-Ming Li, Kun-Yu Lin, Wei-Shi Zheng
   ACM MM 2026
   Abstract: ChoreoPlan bridges reference-dance quality and humanoid executability. It couples beat-snapped hybrid phrase planning with an Embodied Selector trained on offline controller rollouts to rank feasible, semantically aligned motions. On AIST++ and FineDance in IsaacGym and MuJoCo, it improves rollout success, tracking, rhythm, and semantics, with transfer to a Unitree G1.

2. **Beyond Mimicry: Learning Whole-Body Human-Humanoid Interaction from Human-Human Demonstrations**
   Wei-Jin Huang, Yue-Yi Zhang, Yi-Lin Wei, Zhi-Wei Xia, Juantao Tan, Yuan-Ming Li, Zhilin Zhao, Wei-Shi Zheng
   CVPR 2026
   Abstract: Physical human-humanoid interaction is hindered by scarce high-quality data. We leverage abundant human-human data via PAIR, a contact-preserving retargeting pipeline, and D-STAR, a hierarchical interaction policy that attains a state-of-the-art 75.4% average success rate across six interactive tasks and transfers to a real Unitree G1 humanoid.

3. **Modeling Multiple Normal Action Representations for Error Detection in Procedural Tasks**
   Wei-Jin Huang, Yuan-Ming Li, Zhi-Wei Xia, Yu-Ming Tang, Kun-Yu Lin, Jian-Fang Hu, Wei-Shi Zheng
   CVPR 2025
   Abstract: Error detection in procedural activities is essential for AR-assisted and robotic systems, yet prior methods rely on temporal ordering or static prototypes of normal actions. We propose AMNAR, an Adaptive Multiple Normal Action Representation framework that predicts all valid next actions and reconstructs their normal representations for comparison with the ongoing action, setting state-of-the-art across EgoPER, HoloAssist, and CaptainCook4D (+7.4% EDA / +6.5% AUC over the prior best).

4. **Less Static, More Private: Towards Transferable Privacy-Preserving Action Recognition by Generative Decoupled Learning**
   Zhi-Wei Xia, Kun-Yu Lin, Yuan-Ming Li, **Wei-Jin Huang**, Xian-Tuo Tan, Wei-Shi Zheng
   ICCV 2025
   Abstract: Privacy-preserving action recognition aims to protect individual privacy in action videos without compromising recognition, but existing methods struggle under video domain shifts. We propose GenPriv, a transferable framework that decouples static and dynamic video features and removes privacy-sensitive content from static features, achieving strong cross-domain action recognition while preserving privacy.

5. **EgoExo-Fitness: Towards Egocentric and Exocentric Full-Body Action Understanding**
   Yuan-Ming Li, **Wei-Jin Huang (Co-First)**, An-Lan Wang, Ling-An Zeng, Jing-Ke Meng, Wei-Shi Zheng
   ECCV 2024
   Abstract: We present EgoExo-Fitness, a 32-hour full-body action understanding dataset of 1,276 cross-view fitness sequences from 40 subjects, captured by six synchronized egocentric and exocentric cameras. It is richly annotated with two-level temporal boundaries, natural-language comments, and 1-5 action-quality scores, enabling fine-grained interpretable action analysis beyond existing datasets.

Honors and Awards

2024 President Scholarship, Sun Yat-sen University (top 5%)
2022 Kaggle – H&M Personalized Fashion Recommendations, Silver (85/2952)
National Scholarship (1/66) • Tencent Scholarship – First Class (1/66)
2021 Kaggle – G2Net Gravitational Wave Detection, Silver (57/1219)